Set
Fri 11 Sep 2026

Homework 2 — Sessions 3, 4 & 5

Functions · Function families · Proof techniques

Part A · Session 3 — functions, composition and the unit circle

1 IMAGE AND PREIMAGE

Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x^2 - 6x + 5$.

a. Find the image of f and prove your answer. State clearly which part of the argument shows the image is contained in your set, and which shows every element of your set is attained.

b. Compute $f^{-1}(\{0\})$ and $f^{-1}(\{-5\})$. Explain in one sentence what the second answer says about the relationship between the image and the codomain.

> **SOLUTION** (a) Complete the square: $f(x) = (x-3)^2 - 4$. Since $(x-3)^2 \geq 0$ for every real x, we get $f(x) \geq -4$, so the image is contained in $[-4, \infty)$. Conversely, given any $y \geq -4$, put $x = 3 + \sqrt{y+4}$ — which is real because $y + 4 \geq 0$ — and then $f(x) = y$. Hence the image is exactly $[-4, \infty)$. The first half is the containment; the second is attainment.
> (b) $f(x) = 0$ means $(x-3)^2 = 4$, so $x - 3 = \pm 2$ and $f^{-1}(\{0\}) = \{1, 5\}$. For -5: since $f(x) \geq -4 > -5$ for every x, no input maps to -5, so $f^{-1}(\{-5\}) = \varnothing$. An empty preimage is exactly what it means for a codomain element to lie outside the image — the codomain is declared, the image is computed.

2 RESTRICTION AND INVERSE

Keep $f(x) = x^2 - 6x + 5$.

a. Show that $f : \mathbb{R} \to \mathbb{R}$ is not injective, with an explicit witness.

b. Choose a domain D and codomain C making $f : D \to C$ a bijection, and prove *both* halves.

c. Write down the inverse function explicitly, and verify one composite.

SOLUTION (a) $f(1) = 0 = f(5)$ while $1 \neq 5$, so f is not injective. (Any pair symmetric about $x = 3$ works.)
(b) Take $D = [3, \infty)$ and $C = [-4, \infty)$. *Injective:* suppose $a, b \in D$ with $f(a) = f(b)$. Then $(a-3)^2 = (b-3)^2$, so $|a-3| = |b-3|$; as $a, b \geq 3$ both quantities are non-negative, giving $a - 3 = b - 3$ and $a = b$. *Surjective:* given $y \in C$, the value $x = 3 + \sqrt{y+4}$ lies in D (the square root is ≥ 0) and satisfies $f(x) = y$, by part 1(a).
(c) $f^{-1}(y) = 3 + \sqrt{y+4}$ for $y \geq -4$. Check: $f\left(f^{-1}(y)\right) = \left(3 + \sqrt{y+4} - 3\right)^2 - 4 = (y+4) - 4 = y$.

> The choice $D = (-\infty, 3]$ with $f^{-1}(y) = 3 - \sqrt{y+4}$ is equally correct – but the domain must be stated, because "the function $x^2 - 6x + 5$" names no single function.

3 COMPOSITION

Let $g(x) = 2x - 1$ and $h(x) = x^2 + 1$, both from $\mathbb{R}$ to $\mathbb{R}$.

a. Compute $h \circ g$ and $g \circ h$ as explicit formulas, and give a value of x at which they differ. What does this show about composition?

b. Prove that g is a bijection and find g^{-1}.

c. Prove or disprove: $h \circ g$ is injective on $\mathbb{R}$.

SOLUTION (a) $(h \circ g)(x) = (2x-1)^2 + 1 = 4x^2 - 4x + 2$, and $(g \circ h)(x) = 2(x^2+1) - 1 = 2x^2 + 1$. At $x = 1$ the first gives 2 and the second gives 3. So composition is **not commutative**: the order of the maps matters.
(b) *Injective:* $2a - 1 = 2b - 1 \Rightarrow a = b$. *Surjective:* given y, take $x = \frac{y+1}{2}$. So g is a bijection with $g^{-1}(y) = \frac{y+1}{2}$.
(c) **Disproved.** $(h \circ g)(x) = 4x^2 - 4x + 2 = (2x-1)^2 + 1$, so $(h \circ g)(0) = 2 = (h \circ g)(1)$ while $0 \neq 1$. Note g *is* injective – so an injective first map does not make the composite injective. (The implication runs the other way: if $h \circ g$ is injective, then g must be.)

4 UNIT CIRCLE

Derive each value from the unit circle. For each, state the quadrant and the reference angle before giving the number; a recalled value with no working is not a derivation.

a. $\cos\dfrac{5\pi}{6}$

b. $\sin\dfrac{7\pi}{4}$

c. $\tan\dfrac{2\pi}{3}$

d. $\cos\dfrac{11\pi}{6}$

Then explain, in one sentence each, why $\sin$ is not injective on $\mathbb{R}$ and what restriction of its domain makes it so.

> **SOLUTION** (a) Quadrant II, reference $\frac{\pi}{6}$; cosine is the x-coordinate and is negative there, so $-\frac{\sqrt{3}}{2}$.
> (b) Quadrant IV, reference $\frac{\pi}{4}$; sine is the y-coordinate and is negative there, so $-\frac{\sqrt{2}}{2}$.
> (c) Quadrant II, reference $\frac{\pi}{3}$; $\tan = \frac{\sin}{\cos}$ is negative there, so $-\sqrt{3}$.
> (d) Quadrant IV, reference $\frac{\pi}{6}$; cosine is positive there, so $\frac{\sqrt{3}}{2}$.
> $\sin$ is not injective because it is periodic: $\sin(\theta + 2\pi) = \sin\theta$, so for instance $\sin 0 = \sin 2\pi = 0$ with $0 \neq 2\pi$. Restricting the domain to $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ makes it injective (and onto $[-1, 1]$), which is exactly how $\arcsin$ is defined.

5 CLASSIFICATION

For each function, decide whether it is injective, surjective, both, or neither. Prove every positive claim; disprove every negative one with an explicit witness.

a. $f : \mathbb{Z} \to \mathbb{Z},\ \ f(n) = 3n - 7$

b. $f : \mathbb{R} \to \mathbb{R},\ \ f(x) = x^3 + x$

c. $f : \mathbb{R} \setminus \{2\} \to \mathbb{R} \setminus \{1\},\ \ f(x) = \dfrac{x+3}{x-2}$

> **SOLUTION** (a) **Injective, not surjective.** $3a - 7 = 3b - 7 \Rightarrow a = b$. Not onto: $f(n) = 0$ would need $n = \frac{7}{3} \notin \mathbb{Z}$; more simply every output is $\equiv -7 \equiv 2 \pmod 3$, so 0 is never attained.
> (b) **Bijective.** Injective: if $a < b$ then $a^3 < b^3$ and $a < b$, so $f(a) < f(b)$ – a strictly increasing function is injective. Surjective: f is continuous with $f(x) \to \pm\infty$ as $x \to \pm\infty$, so by the intermediate value theorem every real is attained.
> (c) **Bijective.** Injective: $\frac{a+3}{a-2} = \frac{b+3}{b-2}$ cross-multiplies to $(a+3)(b-2) = (b+3)(a-2)$, i.e. $ab - 2a + 3b - 6 = ab - 2b + 3a - 6$, giving $5b = 5a$ and $a = b$. Surjective: solving $y = \frac{x+3}{x-2}$ gives $x = \frac{2y+3}{y-1}$, defined for every $y \neq 1$. This x cannot equal 2, since $\frac{2y+3}{y-1} = 2$ would give $2y + 3 = 2y - 2$. Thus the preimage lies in the stated domain; this is exactly why 1 was removed from the codomain.

6 POLYNOMIAL STRUCTURE

Let $p(x) = 2x^3 - 3x^2 - 11x + 6$.

a. State the degree and leading coefficient, and describe the end behaviour with a reason.

b. Given that $x = 3$ is a root, factor p completely over $\mathbb{R}$ and list all roots.

c. Sketch the graph, marking every x-intercept and the y-intercept.

> **SOLUTION** (a) Degree 3, leading coefficient 2. The degree is odd and the leading coefficient positive, so $p(x) \to -\infty$ as $x \to -\infty$ and $p(x) \to +\infty$ as $x \to +\infty$: down on the left, up on the right. For large $|x|$ the term $2x^3$ dominates the rest.
> (b) Dividing by $(x-3)$ gives $p(x) = (x-3)(2x^2+3x-2) = (x-3)(2x-1)(x+2)$. The roots are $x = 3$, $x = \frac{1}{2}$ and $x = -2$, each of multiplicity one.
> (c) y-intercept $p(0) = 6$. The curve crosses at -2, $\frac{1}{2}$ and 3 — it crosses rather than touches at each, because every factor occurs once.

7 EXPONENTIAL EQUATION

Solve over the reals:

$$3^{2x} - 12 \cdot 3^x + 27 = 0.$$

Justify the substitution you use, and say why *both* resulting values must be checked before you convert back.

> **SOLUTION** Put $u = 3^x$, so $3^{2x} = (3^x)^2 = u^2$ and the equation becomes $u^2 - 12u + 27 = 0$. The substitution is legitimate because 3^x is defined for every real x — but it takes **only positive values**, which is why each root must be tested: a non-positive root would correspond to no real x at all.
> Factoring, $u^2 - 12u + 27 = (u-3)(u-9)$, so $u = 3$ or $u = 9$. Both are positive, so both survive. Since 3^t is one-to-one (base 3 is positive and $\neq 1$),
>
> $$3^x = 3 \Rightarrow x = 1, \qquad 3^x = 9 = 3^2 \Rightarrow x = 2.$$
>
> Check: at $x = 1$, $9 - 36 + 27 = 0$; at $x = 2$, $81 - 108 + 27 = 0$. ✓

8 LOGARITHMIC EQUATION

Solve exactly for x:

$$\log_3(x+6) - \log_3(x-2) = 2.$$

State the domain restriction *before* you combine the logarithms, and check your answer in the original equation.

SOLUTION *Domain first.* Both inputs must be positive: $x+6>0$ and $x-2>0$, so $x>2$ (the stronger condition). Everything below happens on that domain.

By the quotient law, $\log_3\left(\dfrac{x+6}{x-2}\right)=2$, so $\dfrac{x+6}{x-2}=3^2=9$. Then $x+6=9(x-2)=9x-18$, giving $24=8x$ and $x=3$.

$3>2$, so the answer is admissible. Check: $\log_3 9-\log_3 1=2-0=2$. ✓

> Stating the domain is not bookkeeping: combining two logarithms into one *enlarges* the set of x the equation makes sense for, so an answer found after combining can easily be outside the original domain.

9 CHANGE OF BASE

a. Compute $\log_8 32$ and $\log_9 27$ exactly, showing the change-of-base step.

b. Prove that $\log_a b\cdot\log_b a=1$ for all valid bases a,b, and state exactly which conditions on a and b the proof needs.

SOLUTION (a) $\log_8 32=\dfrac{\ln 32}{\ln 8}=\dfrac{5\ln 2}{3\ln 2}=\dfrac{5}{3}$. $\log_9 27=\dfrac{\ln 27}{\ln 9}=\dfrac{3\ln 3}{2\ln 3}=\dfrac{3}{2}$.
Both work because each pair is a power of a common base.

(b) By change of base, $\log_a b=\dfrac{\ln b}{\ln a}$ and $\log_b a=\dfrac{\ln a}{\ln b}$, so the product is $\dfrac{\ln b}{\ln a}\cdot\dfrac{\ln a}{\ln b}=1$. The proof needs $a,b>0$ so the logarithms exist, and $a\neq 1$, $b\neq 1$ so that $\ln a\neq 0$ and $\ln b\neq 0$ — otherwise a denominator vanishes.

10 GROWTH COMPARISON

Consider 2^n and n^3 for positive integers n.

a. Find the smallest n_0 such that $2^n > n^3$ for every $n \geq n_0$. Give the two computations either side of the crossover, then give a short argument that the inequality continues to hold for every later n.

b. Explain why checking a handful of values does not establish the claim for all $n \geq n_0$, and name the proof technique that would.

> **SOLUTION** (a) $n_0 = 10$. At $n = 9$: $2^9 = 512$ while $9^3 = 729$, so the inequality fails. At $n = 10$: $2^{10} = 1024$ while $10^3 = 1000$, so it holds. Beyond $n = 10$ the gap only widens, since doubling outpaces the cube's growth factor $\left(\frac{n+1}{n}\right)^3$, which drops below 2 once $n \geq 4$.
> (b) A finite check covers finitely many n; the claim is about infinitely many. Each verification is one instance of a universally quantified statement, and instances never exhaust a "for every" claim. The technique that does settle it is **induction** – prove the base case $n = 10$, then show $2^k > k^3$ forces $2^{k+1} > (k+1)^3$. That is Session 6.

11 APPLICATION · PERIODIC MODEL

Bangkok's temperature over one day is modelled by

$$T(t) = 8\sin\left(\frac{\pi(t-9)}{12}\right) + 22,$$

with T in °C and t the hour, $0 \leq t \leq 24$.

a. State the amplitude, midline and period, showing how you get the period from the coefficient.

b. Give the maximum and minimum temperatures and the times at which they occur.

c. The model has period 24 hours. What does that say about $T(0)$ and $T(24)$, and is that physically reasonable?

> **SOLUTION** (a) Amplitude $|A| = 8$, midline $y = 22$. The coefficient of t inside the sine is $B = \frac{\pi}{12}$, and the period is $\frac{2\pi}{B} = \frac{2\pi}{\pi/12} = 24$ hours – B sits *inside* the function, so you divide 2π by it rather than multiply.
> (b) Sine reaches 1 when its input is $\frac{\pi}{2}$: $\frac{\pi(t-9)}{12} = \frac{\pi}{2}$ gives $t - 9 = 6$, so $t = 15$ and $T = 8 + 22 = 30$°C. Sine reaches -1 when the input is $-\frac{\pi}{2}$, giving $t = 3$ and $T = 22 - 8 = 14$°C. Maximum 30°C at 15:00, minimum 14°C at 03:00.
> (c) Periodicity forces $T(24) = T(0) = 8\sin(-\frac{3\pi}{4}) + 22 \approx 16.3$°C. That is reasonable as a modelling assumption – midnight tonight should resemble midnight last night – but it is an assumption, not a fact about weather: a real cold front breaks it.

12 CONTRAPOSITIVE

Prove that for every integer n, if n^3 is odd then n is odd.

a. State the contrapositive of the claim.

b. Prove the contrapositive, and say in one sentence why proving it settles the original.

c. Would a direct proof have been harder? Say why.

> **SOLUTION** (a) Contrapositive: for every integer n, if n is **even** then n^3 is **even**.
> (b) Let n be an arbitrary even integer. By definition $n = 2k$ for some $k \in \mathbb{Z}$. Then $n^3 = (2k)^3 = 8k^3 = 2(4k^3)$, and $4k^3 \in \mathbb{Z}$, so n^3 is even. ■ An implication and its contrapositive are true in exactly the same cases, so proving one proves the other.
> (c) Yes. A direct proof starts from "n^3 is odd", i.e. $n^3 = 2m + 1$, and must extract information about n from a statement about its cube — there is no convenient factorisation to use. The contrapositive starts from $n = 2k$, which is a usable formula immediately. That is the general rule: contrapose when the negated conclusion is the more workable hypothesis.

13 CONTRADICTION

Prove that $\sqrt{3}$ is irrational.

a. Write out the proof by contradiction in full.

b. Your argument will use the fact that if $3 \mid a^2$ then $3 \mid a$. State where you used it, and why the analogous step would fail if 3 were replaced by 4.

> **SOLUTION** (a) Suppose $\sqrt{3}$ is rational. Then $\sqrt{3} = \frac{a}{b}$ with $a \in \mathbb{Z}$, $b \in \mathbb{N}$, and a, b sharing no common factor greater than 1. Squaring, $a^2 = 3b^2$, so $3 \mid a^2$ and therefore $3 \mid a$. Write $a = 3a_1$. Substituting, $9a_1^2 = 3b^2$, so $b^2 = 3a_1^2$ and by the same fact $3 \mid b$. But then 3 divides both a and b, contradicting that they share no factor above 1. So $\sqrt{3}$ is irrational. ■
> (b) The fact is used twice: once to get $3 \mid a$ from $3 \mid a^2$, and once to get $3 \mid b$ from $3 \mid b^2$. It holds because 3 is **prime**. For 4 it fails: $4 \mid 2^2$ but $4 \nmid 2$. And indeed the conclusion would be false — $\sqrt{4} = 2$ is rational.

14 COUNTEREXAMPLE

Consider the claim: *for every non-negative integer n, the value $n^2 - n + 41$ is prime.*

a. Verify the claim for $n = 0, 1, 2$ and 3.

b. The claim is nevertheless false. Find a counterexample and show it fails.

c. In two or three sentences, say what this example shows about the relationship between evidence and proof.

> **SOLUTION** (a) $n = 0$: 41. $n = 1$: 41. $n = 2$: 43. $n = 3$: 47. All four are prime.
> (b) Take $n = 41$. Then
>
> $$41^2 - 41 + 41 = 41^2 = 1681 = 41 \times 41,$$
>
> which is composite. So the claim is false, and $n = 41$ is a counterexample.
> (c) The formula is prime for every n from 0 to 40 – forty-one consecutive successes – and still the universal claim is false. No finite amount of confirming evidence establishes a "for every" statement; it only fails to refute it. One counterexample, by contrast, settles the matter permanently. This is why the course insists on proof rather than checking.

15 QUANTIFIERS AND WLOG

a. Write the negation of each statement, in words, without using the phrase "it is not the case that": (i) for every real x, there exists an integer n with $n > x$; (ii) there exists a real x such that for every real y, $xy = y$.

b. Prove: for all real a, b, equality holds in $|a + b| \leq |a| + |b|$ if and only if $ab \geq 0$. You may use "without loss of generality" – if you do, name the symmetry that licenses it.

> **SOLUTION** (a) (i) There exists a real x such that for every integer n, $n \leq x$. (ii) For every real x, there exists a real y with $xy \neq y$. In each case the quantifiers flip and the innermost property is negated.
> (b) Both sides are non-negative, so the equality is equivalent to the equality of their squares:
>
> $$|a+b|^2 = (|a| + |b|)^2 \iff a^2 + 2ab + b^2 = a^2 + 2|a||b| + b^2 \iff ab = |ab|.$$
>
> And $ab = |ab|$ holds exactly when $ab \geq 0$. ■
> A WLOG is available but not needed here. The statement is symmetric in a and b, so one could assume $|a| \geq |b|$ after swapping their names if needed. One could also assume $a \geq 0$: when $a < 0$, replacing (a, b) by $(-a, -b)$ preserves both sides of the equality and the sign of ab. In each case, the named symmetry is what makes the WLOG step valid.